
Privacy Analysis of Augmented Reality Glasses

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Weekly Report 7

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What has been done (Previous Week)

- Successfully created a static analysis of the APK of the Meta View app using MobSF and exported it as a json file.
- Captured Bluetooth traffic for both classic and ble, however, not fully done as I had some questions regarding the captures I did do.
- All captures done has very basic documentation and their respective voice recording uploaded alongside them.
- Did more capturing of BLE traffic during the initial pairing of the glasses and the phone to try and see if I could capture them sending any pairing sequences or key exchanges to use for decrypting the Bluetooth traffic. Was unsuccessful but the captures + basic documentation is uploaded onto gitlab in the folder data/Capturing traffic over BLE/Experiments for key exchanges.
- Finished the documentation for the Bluetooth sniffer, the Problems encountered with the glasses, and Basic Instructions (as of now) and uploaded.
- Added some more documentations in regards to the sections Notes and/or Time stamps for the Wi-Fi experiments, also uploaded.
- Tried to use Android Studio Emulator for the dynamic analysis but was unsuccessful.

What will be done next week

- Do more experiments for Bluetooth traffic capturing
- Continue with editing the documentations for the captures
- Write documentation for the static analysis
- For decryption: try to root the phone as well as learn about and set up mitmproxy
- Try Genymotion for the emulation and have a dynamic analysis of the app
- Go through the description of the thesis and make a small evaluation on what has been done and what needs to be done as well as how to continue for the distant future

Meeting

Had a meeting with Nima on the 8th, where:

- Discussed about the rooting of the phones, ended up receiving a new phone
- Discussed about continuing the app analysis to have a dynamic analysis
- Briefed about mitmproxy
- Asked about the goals and what has been achieved so far